

Cubic Integration of Well-Rounded Logconcave Functions: A Later-Literature Answer to arXiv:1409.6011

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Status

Literature already answered (affirmative), with limited scope. Conclusion item 2 of Cousins–Vempala is answered by Theorem 1.8 of Kook–Vempala. The separate cubic arbitrary-body rounding question in Conclusion item 4 remains open.

Source question

Ben Cousins and Santosh Vempala, arXiv:1409.6011, ask in Conclusion item 2 on PDF page 35:

It would be interesting to extend our algorithm to integrating any well-rounded logconcave function; we expect this should be possible with essentially the same complexity.

The preceding item records the relevant dimension complexity for their well-rounded convex-body volume algorithm as $O^*(n^3)$.

Explicit answering theorem

Yunbum Kook and Santosh S. Vempala, arXiv:2411.13462, explicitly identify this provenance: in their technical overview they write that Cousins–Vempala raised the question of extending the annealing strategy to arbitrary logconcave distributions, and their Result 4 says that they extend the Cousins–Vempala annealing approach to general logconcave integration.

Their Theorem 1.8 on PDF page 7 states that, for every $\varepsilon > 0$ and every integrable well-rounded logconcave function $f = e^{-V} : \mathbb{R}^n \rightarrow \mathbb{R}$ presented by the specified evaluation oracle, a randomized algorithm returns a $(1 + \varepsilon)$ -multiplicative approximation to $\int f$ with probability at least $3/4$, using

$$\tilde{O}\left(\frac{n^3}{\varepsilon^2}\right)$$

evaluation queries. The theorem also gives $\tilde{O}(n^{3.5} \text{polylog } R + n^3/\varepsilon^2)$ for an arbitrary, not initially well-rounded, logconcave function.

This is exactly the requested extension in the well-rounded case: the dimension exponent is cubic, with the usual accuracy dependence made explicit. Because the answering authors cite and describe the original question, the relationship is an explicit literature answer rather than an agent-inferred theorem match.

Scope limitation

This packet closes only Conclusion item 2. Conclusion item 4 asks for an $O^*(n^3)$ rounding algorithm for arbitrary convex bodies. The current 2026 revision of Jia–Laddha–Lee–Vempala, arXiv:2008.02146, gives $\tilde{O}(n^{3.5})$ with the current KLS bound, and the 2026 survey section of Kook–Vempala, arXiv:2507.18021, still calls this the best known general rounding complexity. No cubic general rounding algorithm is claimed here.

Search evidence

The identification was checked against the source text of arXiv:1409.6011, the current source of arXiv:2008.02146, the source and official PDF of arXiv:2411.13462, and the 2026 discussion in arXiv:2507.18021. The exact answering statement is Theorem 1.8 of arXiv:2411.13462; its overview explicitly cites the source work and describes the extension.

References

- [1] B. Cousins and S. S. Vempala, *Gaussian Cooling and $O^*(n^3)$ Algorithms for Volume and Gaussian Volume*, arXiv:1409.6011.
- [2] Y. Kook and S. S. Vempala, *Sampling and Integration of Logconcave Functions by Algorithmic Diffusion*, arXiv:2411.13462, Theorem 1.8.
- [3] H. Jia, A. Laddha, Y. T. Lee, and S. S. Vempala, *Reducing Isotropy and Volume to KLS: Faster Rounding and Volume Algorithms*, arXiv:2008.02146.